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## Letters

# Single-fraction Radiotherapy Should be the Standard of Care for Palliation of Cancer Symptoms in Patients with Limited Life Expectancy

*Madam* — Single-fraction radiotherapy has replaced multi-fraction radiotherapy as the standard of care for the palliation of pain from bone metastases [1]. Multi-fraction radiotherapy is still often used to palliate symptoms from other disease sites. As nearly half of UK radiotherapy activity is with palliative intent, fractionated treatment places a significant financial burden on highly stretched National Health Service finances [2].

Furthermore, multi-fraction palliative radiotherapy also physically and financially burdens terminally ill patients with multiple journeys to and from hospital. We explored whether there was any survival benefit for multi-fraction palliative radiotherapy when compared with single-fraction radiotherapy.

We analysed an unselected group of 233 patients with all tumour types that were treated palliatively, for any indication, in a single cancer centre over a 3 month period. Most were not suitable for palliative chemotherapy. Patient characteristics, such as median age, total radiotherapy dose and number of fractions, are summarised in Table 1. The dose per fraction ranged from 1.8 to 10 Gy. The overall median survival is comparable with other UK centres. The median overall survival was not different between the two groups ($P = 0.3041$).

On survival analysis by Cox regression, the primary tumour site and histology were predictive of overall survival; but the total radiation dose and dose per fraction were not. The hazard ratios for dose, number of fractions and dose per fraction were 0.956, 1.004 and 1.072, respectively.

The lack of survival advantage for multi-fraction palliative radiotherapy suggests that prognosis is influenced by the biology of disease rather than radiation treatment dose or fractions.

Palliative radiotherapy is becoming more technically complex and costly. The National Health Service faces an unprecedented financial crisis and it is critical that resources are used in a cost-effective manner. We propose that single-fraction radiotherapy should be the standard of care for palliation of cancer symptoms from all organs, particularly in patients with a limited life expectancy [3–5].

**Table 1**
Comparison of single-fraction and multi-fraction palliative radiotherapy

| | Single-fraction radiotherapy | Multi-fraction radiotherapy | |
|---|---|---|---|
| Median age | 69.0 years | 67.8 years | $P > 0.17$ (ns)* |
| Median total dose (range) | 8 Gy (8–10 Gy) | 20 Gy (9–45 Gy) | $P < 0.001$* |
| Median fractions (range) | 1 fraction (1) | 5 fractions (3–25) | $P < 0.001$* |
| 30 day mortality | 16.9% | 9.0% | $P = 0.074$ (ns)† |
| Median survival (95% confidence interval) | 5.01 months (3.15, 6.88) | 6.15 months (4.60, 7.69) | $P = 0.3041$ (ns)‡ |
| 12 month survival | 24.7% | 29.6 % | $P > 0.3$ (ns) |

* Mann-Whitney test.
† Pearson chi-squared test.
‡ Log-rank test.

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## References

[1] Chow E, Zeng L, Salvo N, Dennis K, Tsao M, Lutz S. Update on the systematic review of palliative radiotherapy trials for bone metastases. *Clin Oncol* 2012;24:112–124. http://dx.doi.org/10.1016/j.clon.2011.11.004.

[2] Hoskin PJ, Forbes H, Ball C, Riley D, Cooper T, *et al*, Radiotherapy Clinical Information Group. Variations in radiotherapy delivery in England – evidence from the national radiotherapy dataset. *Clin Oncol* 2013;25:531–537. http://dx.doi.org/10.1016/j.clon.2013.04.006.

[3] Ray KJ, Sibson NR, Kiltie AE. Treatment of breast and prostate cancer by hypofractionated radiotherapy: potential risks and benefits. *Clin Oncol* 2015;27:420–426. http://dx.doi.org/10.1016/j.clon.2015.02.008.

[4] Williams M, Woolf D, Dickson J, Hughes R, Maher J, Mount Vernon Cancer Centre. Routine clinical data predict survival after palliative radiotherapy: an opportunity to improve end of life care. *Clin Oncol* 2013;25:668–673. http://dx.doi.org/10.1016/j.clon.2013.06.003.

[5] Siva S, Kirby K, Caine H, Pham D, Kron T, Te Marvelde L, *et al*. Comparison of single-fraction and multi-fraction stereotactic radiotherapy for patients with $^{18}$F-fluorodeoxyglucose positron emission tomography-staged pulmonary oligometastases. *Clin Oncol* 2015;27:353–361. http://dx.doi.org/10.1016/j.clon.2015.01.004.

http://dx.doi.org/10.1016/j.clon.2017.05.009



# Postgraduate Training in a Low- and Middle-income Country: Sharing Experience from a Joint Commission International Accreditation-accredited University Hospital

*Madam* — Our team of residents and faculty of radiation oncology, Aga Khan University, read the article published in your journal with interest [1].

Pakistan, like other low- and middle-income (LMI) countries, needs modern equipment and trained professionals [2].

We are working in a tertiary referral university hospital and our programme is regularly reviewed by internal and external audit systems, including the Joint Commission International Accreditation (JCIA). We are continuously taking steps through audits for training evaluation [3].

Radiation oncology is a quality conscious discipline. Doctors, physicists and radiation therapy technologists (RTT) are integral and crucial components of this service. However, like many countries, in Pakistan they are not getting the deserved recognition as health care providers.

In the absence of robust regulatory systems pertaining to continuing medical education and practice privilege documentation, we strongly recommend peer-reviewed, site-specialised practice [4].

We involve our postgraduates in quality initiatives, like clinical quality indicators, and they are active participants of site-specific tumour boards [5]. We believe and work on pre-defined learning objectives, which are helping us to achieve international postgraduate training standards. We have developed our syllabus as recommended by the International Atomic Energy Agency for LMI countries [6]. We are carrying out end of term briefing sessions to review the progress of structured training.

In order to overcome a shortage of qualified teachers in our country, monthly activities are organised for all trainees in the city, in which learning objective-based interactive learning sessions are conducted by qualified faculty of radiation (oncologist/physicist) on a voluntary basis. This strategy is also helping us to develop collegiality among colleagues of other cancer units. Regular web-based sessions from qualified foreign consultants are also part of this activity. We have also developed training programmes for the training of physicists and RTTs.

In our opinion, radiation oncology is mainly based on the expertise of three personnel: oncologists, physicists and RTTs. Although a lack of equipment is evident, without the expertise of these three professionals, even the availability of machines cannot benefit the patients. The inclusion of quality and ethical practice should be considered as an integral part of structured curriculum. Thus, training is of critical importance and constant efforts are required in order to improve cancer services in LMI countries.

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## References

[1] Eriksen JG. Postgraduate education in radiation oncology in low- and middle-income countries. *Clin Oncol* 2017;29(2):129–134.

[2] Nadeem Abbasi A, Muhammad W, Hussain A. Implementation of quality medical physics training in a low-middle income country—sharing experience from a tertiary care JCIA-accredited university hospital. *J Appl Clin Med Phys* 2016;17(6):454–456.

[3] http://www.jointcommissioninternational.org/about-jci/jci-accredited-organizations/?c=Pakistan.